Anatomy of a Software House

Lecture 6

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4.1 The company

4.2 Company structure

Company

↓

Managing Director

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Financial | **Operations** | Sales & Marketing | **Technical** | Overseas

Operations Director

The Operations Director is responsible for all

- the revenue earning
- operations of the company
- all projects are completed satisfactorily
- Staff is at satisfactory level
- Conflicts may rise to maximize staff utilization and revenue
  - Individual wants to learn other skills
  - Wants individual to work Abroad
- short-term objectives and the longer term goals of the company
- “clashing structures”
- matching people to projects in an optimal way

Technical Director

The Technical Director is responsible for:

- quality management;
- research and development;
- marketing at a technical level (e.g. arranging for staff to give papers at conferences);
- technical training (as opposed to training in, say, project management or presentational skills, which are the responsibility of the personnel function).

4.3 Management of staff

- Recruiting and retain Staff
- Reputation in the market place depends on the quality of its staff
- Loyal Staff
- Training courses provided by other organizations
- Clique-outsiders (a narrow exclusive circle or group of persons especially)
- Staff appraisal
- Employees' achievements and contributions to the company were properly recorded;
- Staff knew what was expected of them and what they needed to achieve in order to gain promotion;
- Proper plans for training and career development were made and regularly reviewed;
- Employees were aware of the company's opinion of their performance.

Going on Problems

Difficult problems to be overcome:

- How to provide continuity of management as staff move from project to project;
- How to use group expertise (e.g. the collective expertise of the company's six database experts) in an environment in which the members of the group are usually working on different projects;
- How to maintain standards and company loyalty as the company continues to grow.

4.4 Producing the budget

- Technical or *revenue earning* staff and *nonrevenue earning* staff
  - different grades have different training needs;
  - senior grades on non-revenue earning activities such as sales support;
  - senior technical staff are usually more specialized and it therefore takes longer to re-deploy them once they have completed a project.

4.4 Producing the budget

- The **sales** staff's estimates of how much more business they could sell, and at what rates, if the staff to handle the business were available;
- The **personnel department's** estimates of the company's ability to recruit new staff of a suitable caliber and to retain existing staff;
- The personnel department's estimate of salary trends in the industry;
- The **operations director's** view of the staff utilization rates achievable;
- **Specific expansion plans**, e.g. plans to open an office in a new location;
- **The financial director's** estimate of the company's ability to raise funds for expansion.

Table 4.1 1999 staff costs and revenue figures (£).

Grade	*Number*	*Annual cost*	*Days worked*	*Utilization (percent)*	*Daily cost*	*Charge rate*	*Total cost*	*Total revenue*
1	20	18,500	195	87	94.87	195	370,000	760,500
2	25	23,000	205	92	112.20	220	575,000	1,127,500
3	42	27,000	206	92	131.07	255	1,134,000	2,206,260
4	41	30,500	204	91	149.51	305	1,250,500	2,551,020
5	27	36,600	198	88	184.85	370	988,200	1,978,020
6	20	39,800	185	83	215.14	440	796,000	1,628,000
7	15	44,000	160	71	275.00	560	660,000	1,344,000
Totals	190			88(avg)			5,773,700	11,595,300

	1999	*2000*
Rent and rates	700,000	720,000
Building maintenance	84,000	88,200
Heating and lighting etc.	40,000	42,000
Cleaning	20,000	21,000
Telephone and postage	85,000	108,039
Equipment depreciation	20,000	25,421
Equipment maintenance	12,000	15,253
Directors' salaries and expenses	480,000	504,000
Management salaries and expenses	945,000	1,201,145
Sales and marketing salaries and expenses	900,000	1,143,947
Secretarial	190,000	241,500
Library and information services	32,000	40,674
Training courses	180,000	228,789
Insurance	25,000	31,776
Legal expenses	20,000	25,421
Interest	400,000	600,000
Total	4,133,000	5,037,166

4.5 Monitoring financial performance

- Each month, the income and expenditure under the various heads are compared

- if significant deviations are observed, corrective action is taken

4.5.1 Costs and revenue

- The budget assumes that, over the twelve month period, staff will increase by a total of 30
- Large projects will cause significant deviations from the budget in various ways.

4.5.2 Project costing

- The costs and revenue of each project are calculated each month
- i.e. the difference between total costs and total revenue to date
- calculated as a percentage of the total revenue.

4.5.3 Sales

- The *confirmed sales report* shows, for each grade, the number of staff in that grade who are committed to contracts in each of the following twelve months and the total expected revenue from that grade in each month.
- The *sales prospects report* shows, for each sales prospect, the potential value of the sale, its likelihood and the likely start date.

4.6 Long-term planning

- The first is to identify appropriate long-term goals
- the second is to identify and formulate plans to overcome structural problems

4.6.1 Expansion plans

- Expansion on the scale desired depends on the ability to recruit and retain suitable staff
- the ability to raise the necessary funding
- the ability to run a much larger and multinational organization.

4.6.2 Company image

- providing high value services at a slight premium level

- The company image of a software house represents how clients, partners, and the public perceive the organization. It is built through professionalism, quality of work, communication style, branding, and overall reputation in the IT industry. A strong company image helps a software house gain client trust, attract skilled employees, and compete effectively in the market.

- A positive image is developed by consistently delivering high-quality software solutions on time and within budget. Meeting client expectations, maintaining transparency, and providing reliable post-deployment support strengthen credibility. Satisfied clients often become long-term partners and provide referrals, further enhancing the company’s reputation.

4.6.2 Company image

- Brand identity also plays a major role. A professional website, strong online presence, active social media profiles, and a clear mission statement reflect seriousness and innovation. Certifications, partnerships, and a well-maintained portfolio of successful projects further improve the company’s public perception.
- Internal culture contributes to company image as well. A software house that values teamwork, continuous learning, ethical practices, and innovation builds a strong internal environment, which reflects externally through quality service and employee satisfaction.

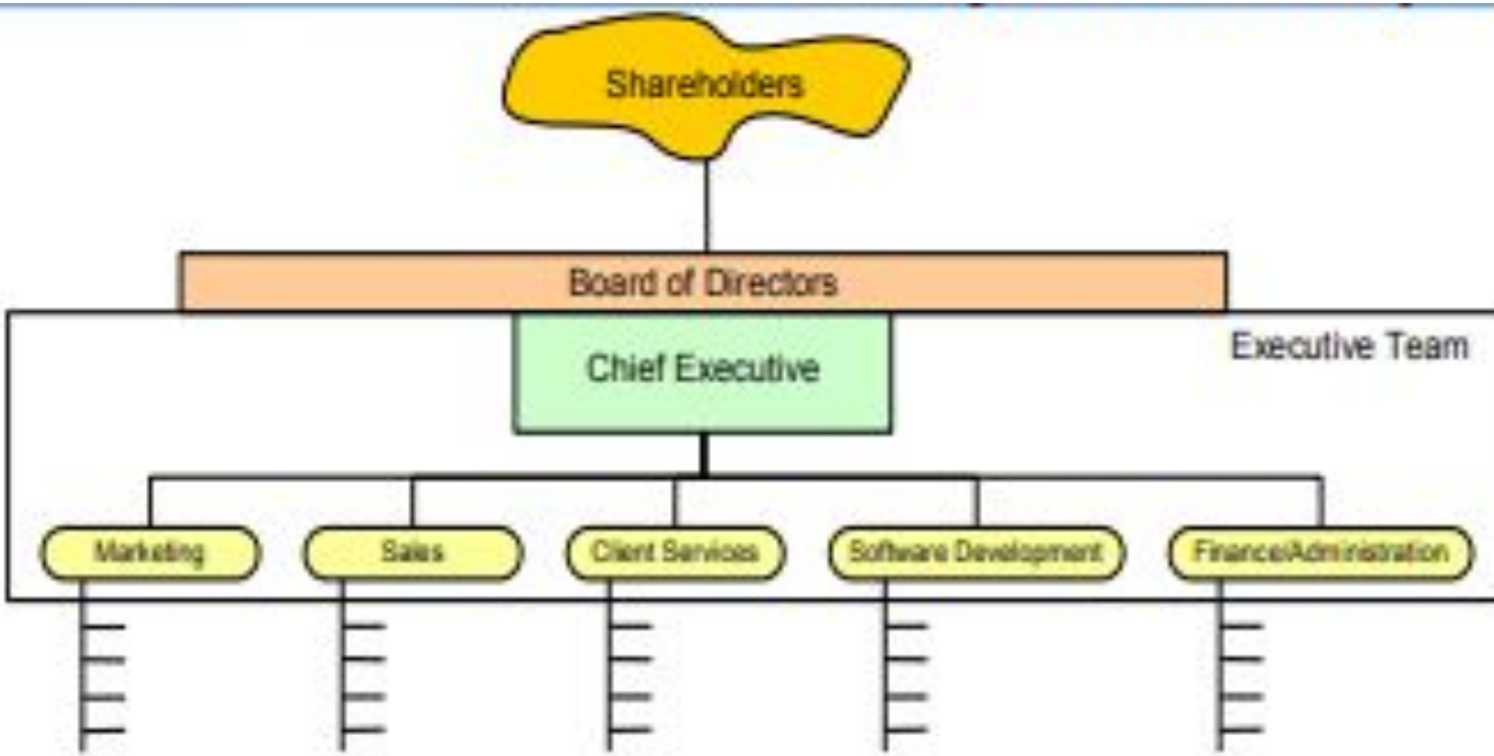
4.6.2 Company image

- In summary, the company image of a software house is a combination of technical excellence, professional behavior, strong branding, and positive client relationships. Maintaining a good image is essential for long-term growth and sustainability in the competitive IT industry.

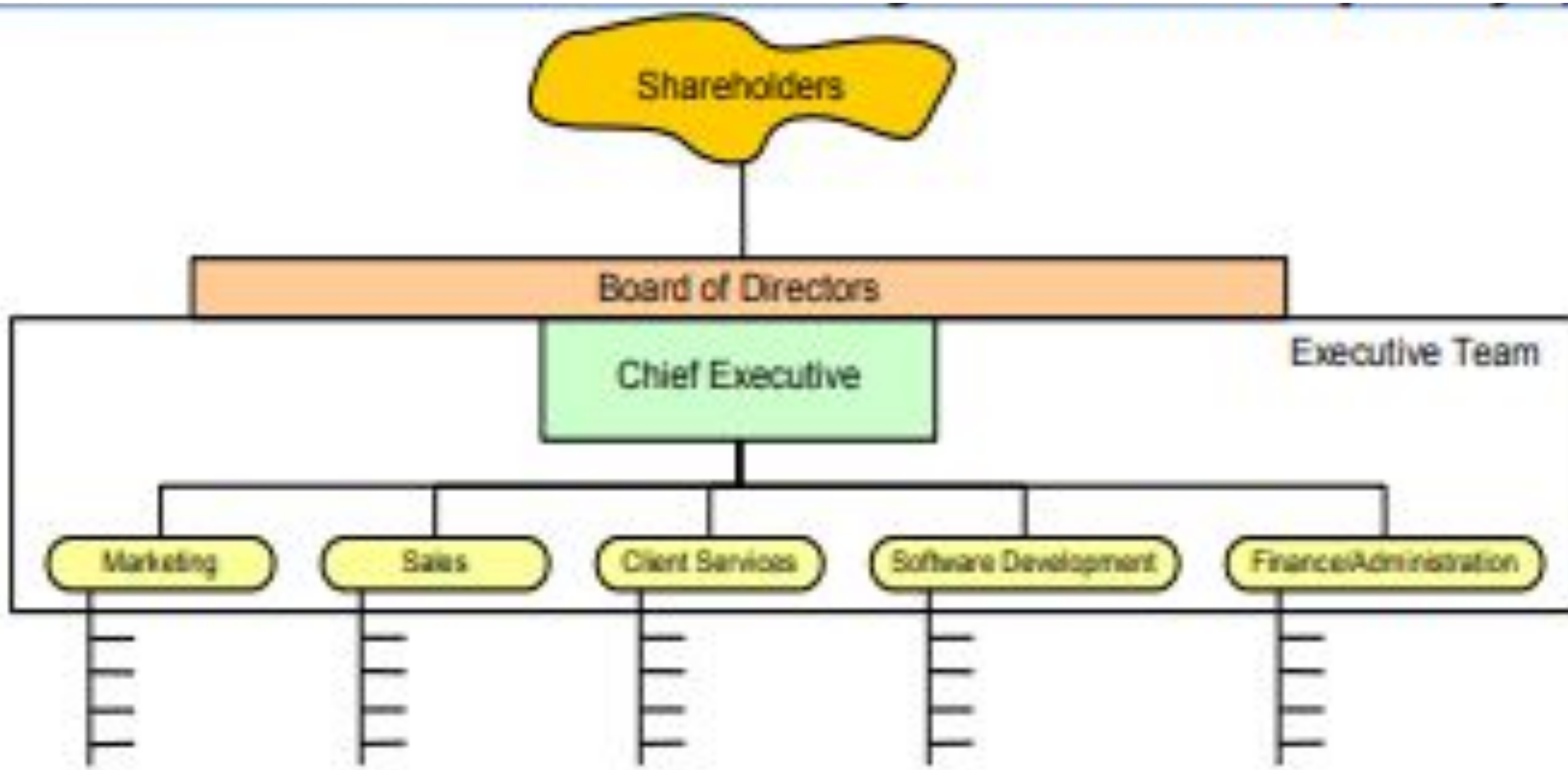
4.6.3.Finance

- under-capitalization that is, there is not enough money invested in the company;

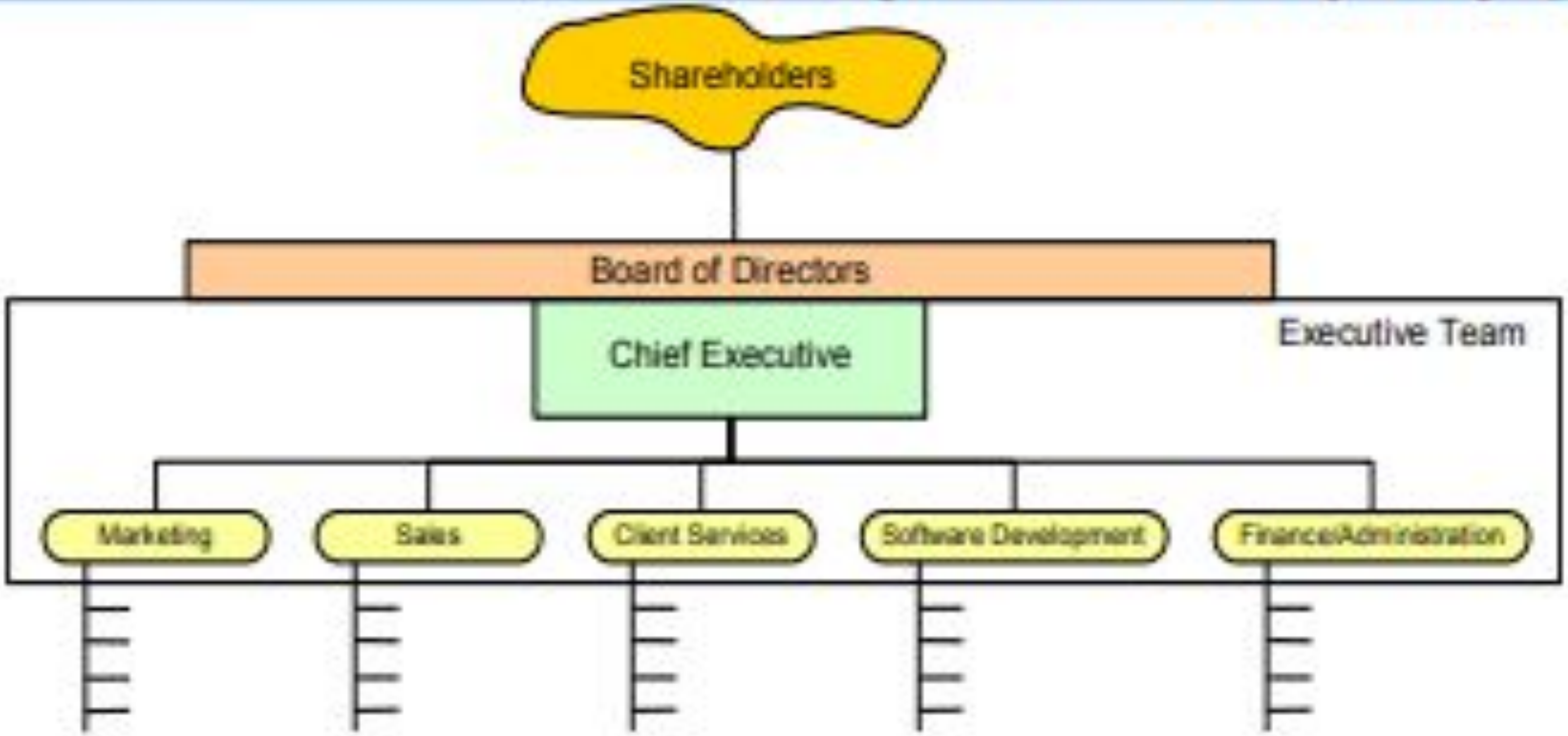
- This means that it must be funded from profits, from loans secured against the personal property of the directors, or from new equity capital;



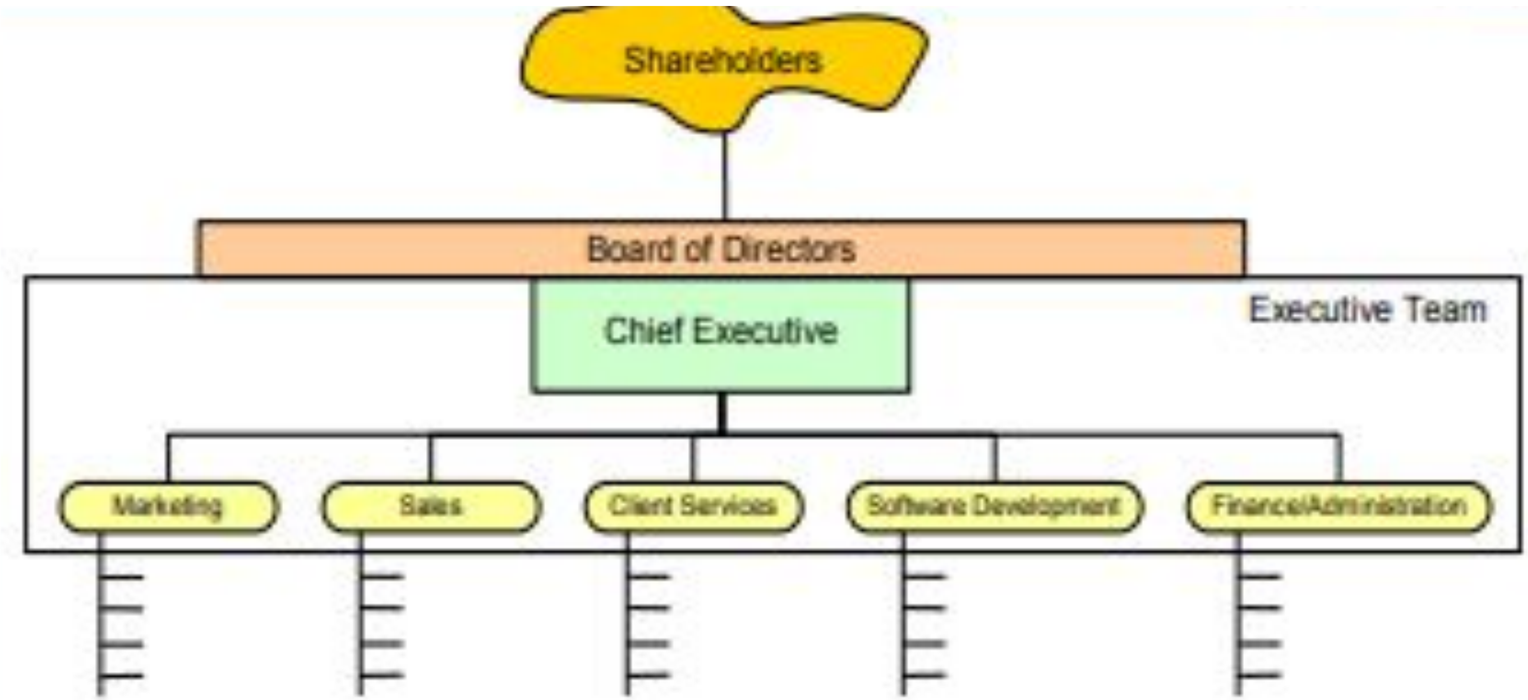
- shareholders
  - owners of the company
  - elect the board of directors, vote on issues
  - same for private & public companies



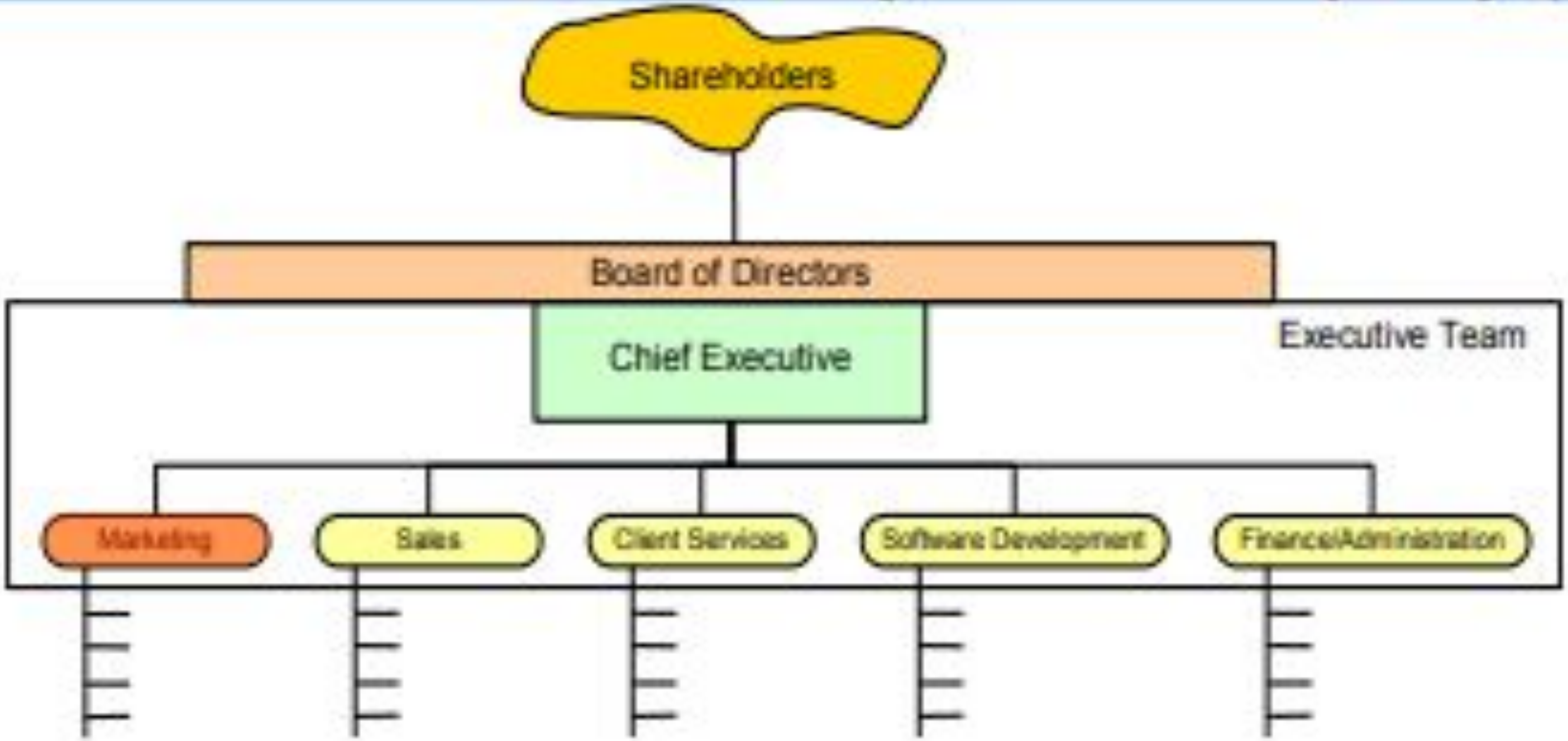
- board of directors
  - represent shareholder interests
  - appoint officers, hire & advise CEO
  - legal liability (or LLC/LTD/etc.)



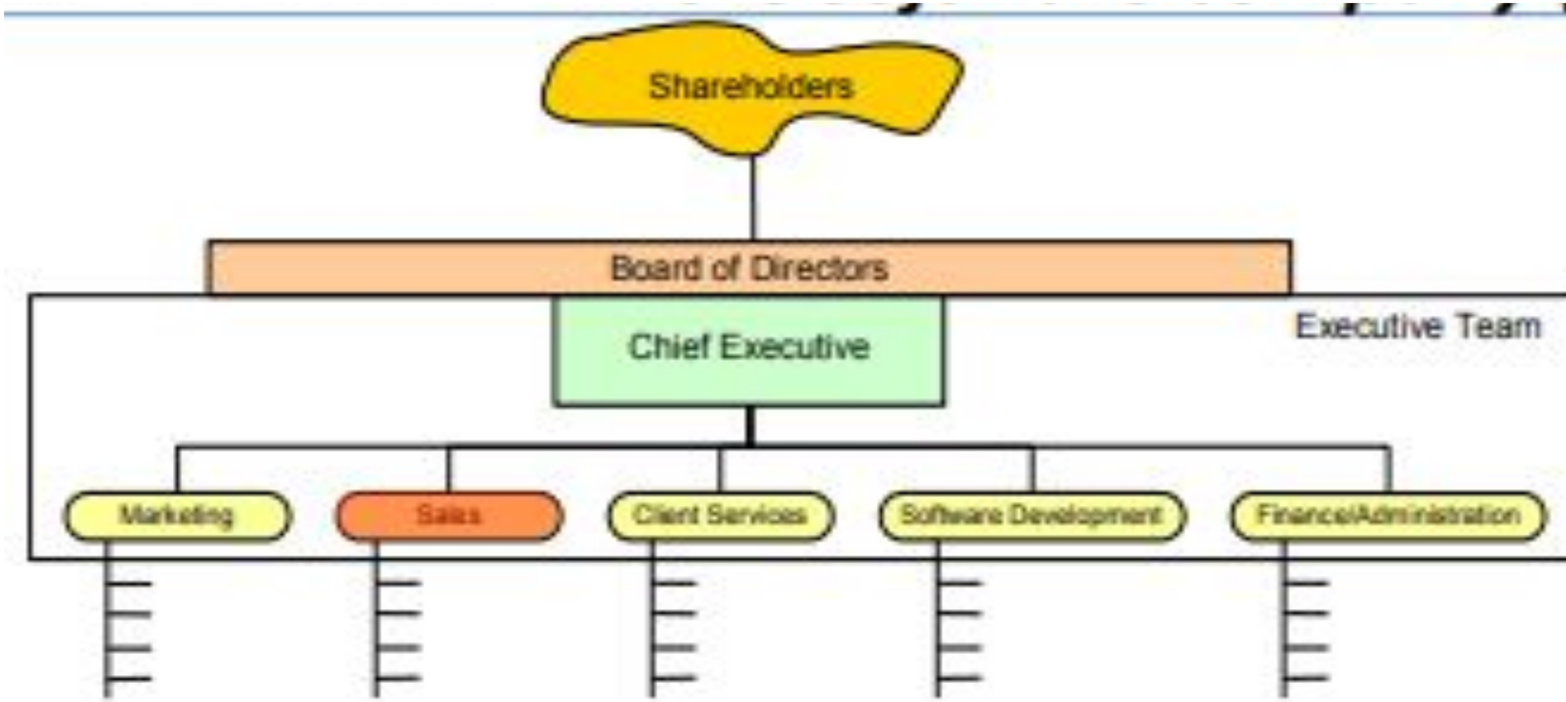
- Chief executive officer (CEO)
  - appointed (or hired) by the board
  - in charge of day-to-day operations & exec. team
  - commits to financial targets (revenue/profit)



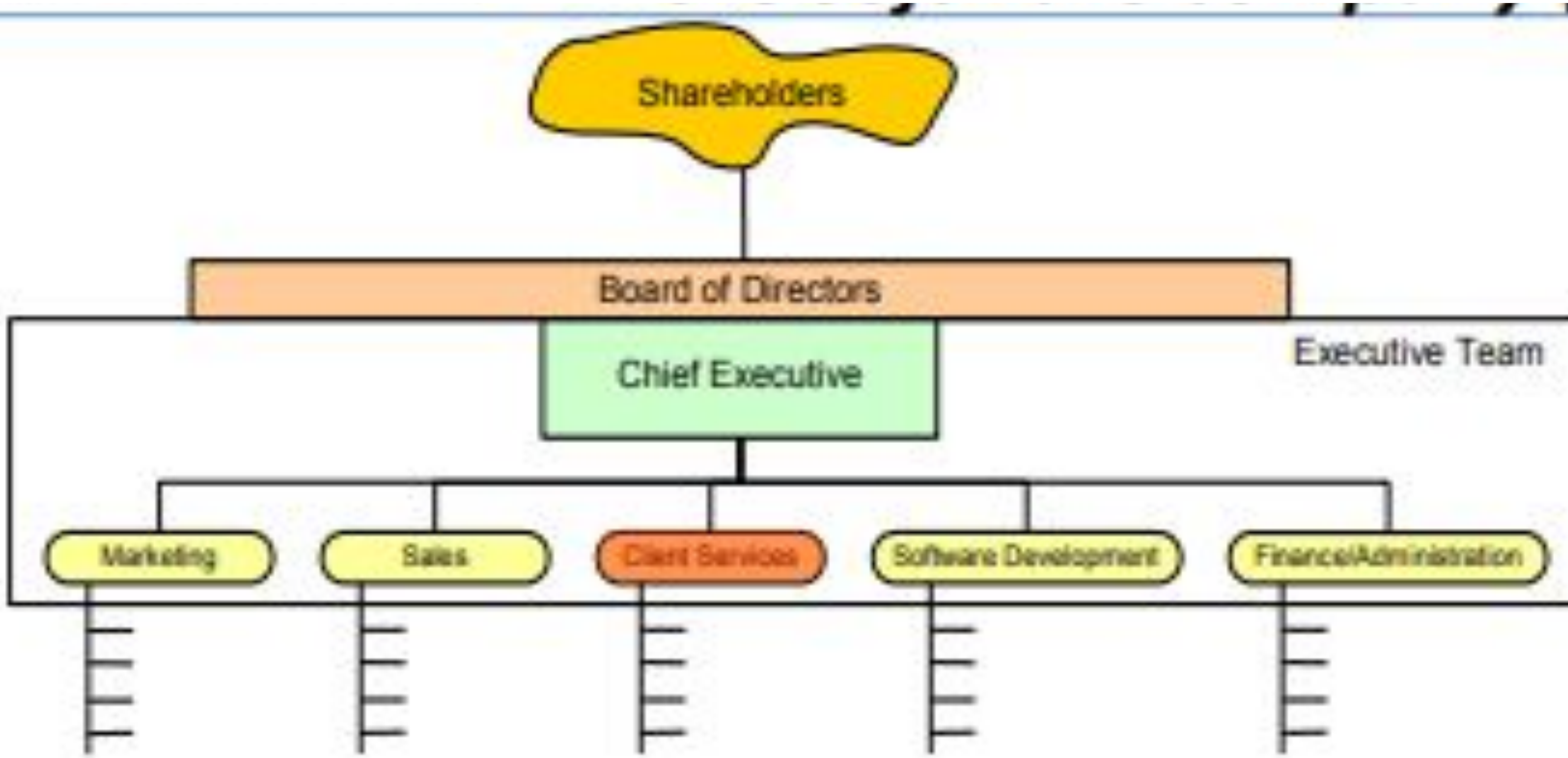
- executive team
  - vp & c-level officers. assembled/managed by CEO
  - in charge of day-to-day functional areas
  - meet regularly to coordinate strategy, budget, etc.



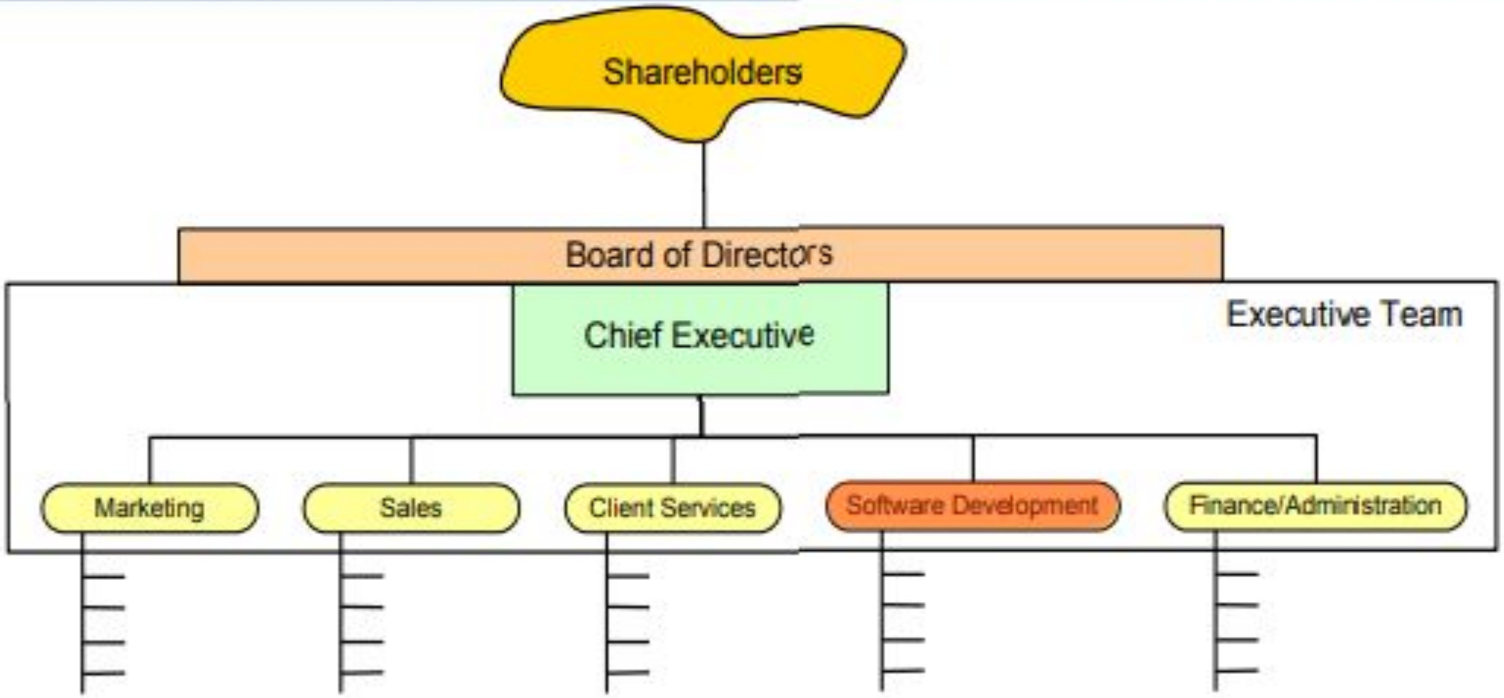
- marketing
  - product mgmt. (sometimes under r&d)
  - marketing communications (MARCOM)
  - business development (BIZDEV)



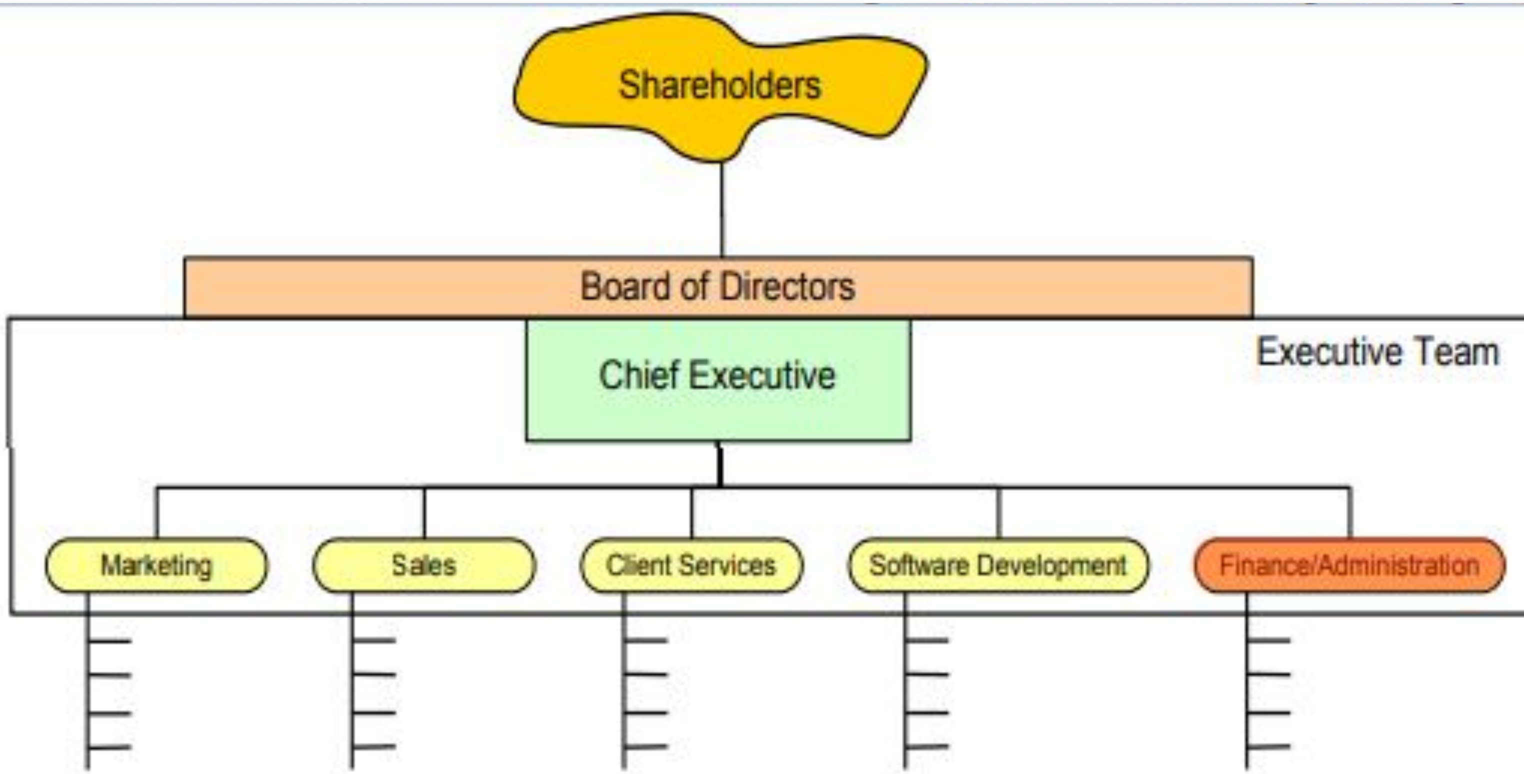
- sales
  - responsible for hitting revenue targets
  - models: direct, dialing-for-dollars, channels, …
  - compensation = base + commission (+ bonus)
  - pipeline: leads, qualified leads, negotiating, close



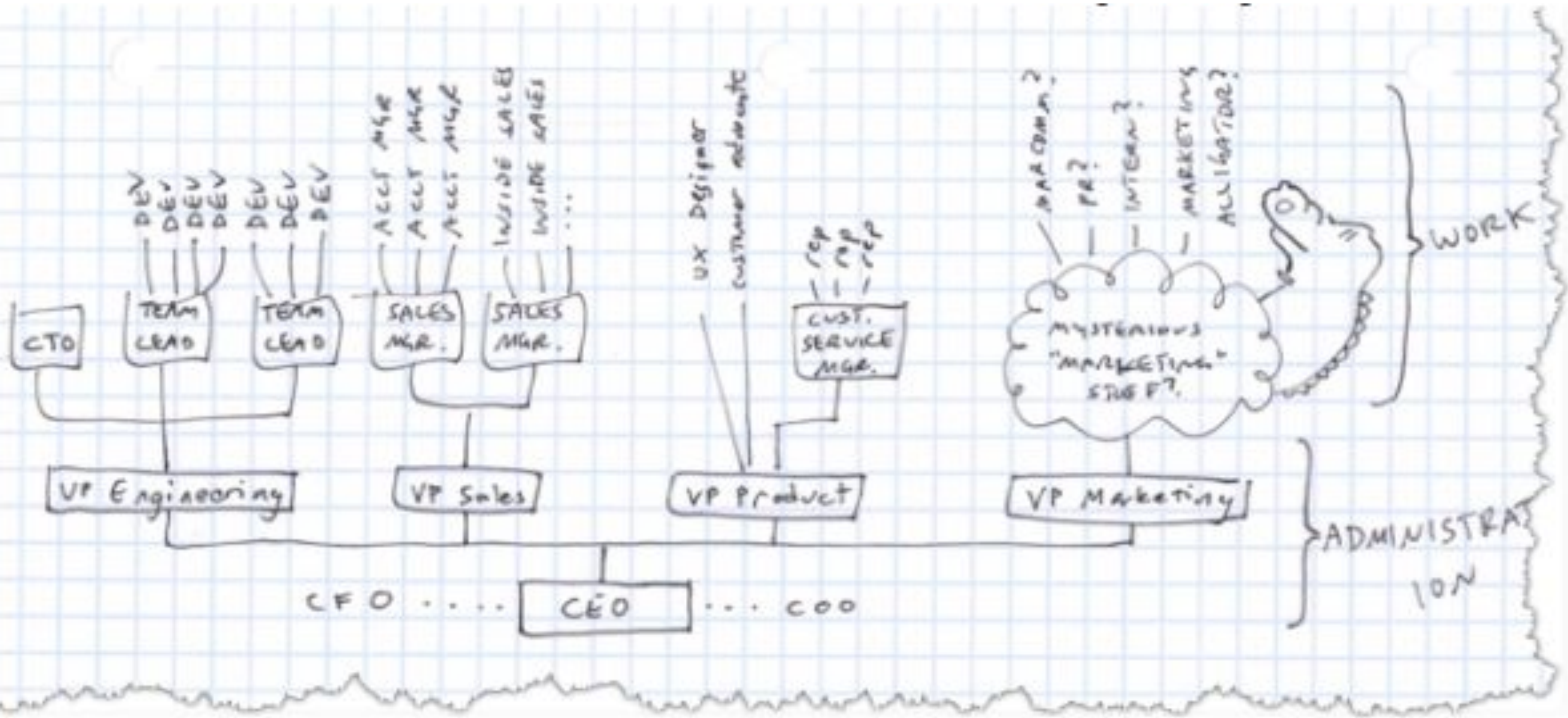
- customer/client services
  - pre-sales support & project planning, deployment
  - account management (ongoing)
  - help desk



- software development (or, r&d)
  - responsible for delivering quality software, solving the correct problems, on time, and with low defect counts
  - CTO vs. VP R&D vs. CSA – who's in charge?



- finance & admin
  - sometimes separate, sometimes include HR
  - investments, payroll, taxes, investor relations, HR, office mgmt., IT, etc.



- this view helps eliminate “command & control”
- hire the best, and then control them, no good!
- this model = more of a “support” structure
- maybe call it “administration?”

Conclusion: Anatomy of a Software House

- In conclusion, the anatomy of a software house reflects a well-structured organization where multiple departments collaborate to design, develop, test, and maintain software products. A successful software house typically consists of key components such as project management, software development, quality assurance, UI/UX design, DevOps, marketing, and client support. Each department plays a specific role, but all work together toward a common goal: delivering high-quality, reliable, and user-centered software solutions.

Conclusion: Anatomy of a Software House

- Effective communication, clear requirement analysis, proper documentation, and the use of structured development methodologies (such as Agile or Waterfall) are essential for smooth operations. Leadership and management ensure that projects are completed on time, within budget, and according to client expectations. Meanwhile, technical teams focus on innovation, coding standards, testing, and system maintenance to maintain product quality.

Conclusion: Anatomy of a Software House

- Ultimately, the strength of a software house lies not only in technical expertise but also in teamwork, adaptability, and continuous improvement. When all components function in harmony, the software house can successfully compete in the fast-growing IT industry and consistently deliver value to its clients.